

Day 9 — Mixed Practice + Module 7 Crossover

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: solve problems combining $\times 11$ and near-base methods.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilam* (IAST: Nikhilam; lit. “the Nikhilam sūtra (also used in Module 9 for subtraction)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Crossover with Module 7.** Each student solves 5 problems combining $\times 11$ method + Nikhilam:
 - $11 \times 99 = 1089$ (use $\times 11$)
 - $97 \times 96 = 9312$ (use Nikhilam)
 - $11 \times 98 = 1078$ (use $\times 11$)
 - $105 \times 95 = 9975$ (use difference-of-squares or Nikhilam mixed)
 - $11 \times 87 = 957$ (use $\times 11$ with carry)
2. **Strategy reflection.** Pair-discuss: “*For each, what method DID you use? Was it the fastest?*”
3. **Build a personal toolkit.** Each student updates their “personal mental-math toolkit” page (started in Module 7) to include Nikhilam multiplication.
4. **One more example.** Solve 998×997 mentally. (Nikhilam: deficits 2, 3 $\rightarrow$ left = 995, right = 006 $\rightarrow$ **995006**.) 4 seconds with the trick; >1 minute by long multiplication.
5. **Reflection.** “*What’s the most useful thing you learned this module?*”

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Mixed Practice + Module 7 Crossover.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).